

Technical Appendix — ED Workflow Coordination

ED Workflow Coordination — Technical Appendix (2025-08-16 14:35 UTC)

Versions & files

- Base: `**ED_Pipeline_v4_KaggleFix.ipynb**`
- Current: `**ED_Pipeline_v4.1.ipynb**` (domain patches: risk scores, lingering, analytics, anti-spam, fixed follow-ups, audit)
- Artifacts: ``metrics.json``, ``emissions.json`` (acceptance, fatigue, suggestions breakdown)

Architecture (small models / skills)

- `**WorkflowState**` (+ `PatientStateManager`): per-encounter, chronological updates; no global state bleed.
- `**Skills**`: tiny, testable units emitting suggestions (no DX). Examples:
 - ``skill_follow_up_imaging_overdue`` (CT > 60m)
 - ``skill_follow_up_cath_activation``, ``skill_follow_up_endo_activation`` (30m)
 - ``skill_troponin_ecg_compliance``, ``skill_second_ecg_with_next_troponin``
 - ``skill_review_troponin_trend`` (display-only delta awareness)
 - ``skill_flag_lingering_patients``, ``skill_offer_snacks``
 - ``skill_surface_relevant_sops``
- `**ResourceTracker**` in core: QR → location/status; reservation TTL; cleaning/maintenance; docking/charge; analytics hook.
- `**Governance & safety**`:
- `**Suggestions-only**`, no auto-orders.
- `**Anti-spam gate**` (cooldown, repeat cap, snooze after accept).
- `**Ethics**`: ``CONFIG["ethics"]["age_sex_in_decisions"]=False`` (scores are display-only).
- `**Learning**`: "shadow" mode (no online parameter changes).

Domain features restored

- QR equipment flow; SOP surfacing; troponin serial + second ECG prompt; CT 60m, Cath/Endo 30m follow-ups; lingering/snacks; language-barrier phrase-card hooks.
- Display-only risk scores: `**MEWS, qSOFA, SOFA-lite, HEART, Marburg, GRACE-proxy**`.

Key fixes today

- `**Kernel/CONFIG/path**` bootstraps for Kaggle (``/kaggle/working``).
- `**NameError/attribute**` cleanup (removed stale refs like ``TinyCritics.benefit_bias``).
- `**Anti-spam**` to stop duplicate CT nags; removed incorrect night CT assumption.
- `**Core wiring**` for equipment + SOP + troponin delta; `**Audit cell**` to prevent regressions.

CONFIG essentials (defaults)

```
CONFIG = {
  "seed": 42,
  "ethics": {"age_sex_in_decisions": False},
  "monitoring": {"lingering_hours": 2.0, "snacks_hours": 4.0},
  "anti_spam": {"cooldown_min": 60, "max_repeats_per_enc": 2, "snooze_after_accept_min": 180},
  "services_mobilization": {"cath_minutes": 30.0, "endo_minutes": 30.0},
  "paths": {"DATA_ROOT": "/kaggle/working", "UI_STATE_DIR": ".../ui", "SOP_DIR": ".../sops",
}
```

Data & validation (to harden next)

- Enforce chronological event order, detect duplicates, validate clinical ranges, require fields per event type.
- Sanitize `event\_attrs` with schema; no silent JSON failures.

Testing targets

- **Unit:** PatientStateManager transitions; ResourceTracker conflicts/TTL; SafetyPolicy; skill outputs for synthetic scenarios.
- **Property tests:** anti-spam invariants ($\leq$ cap per encounter, cooldown respected).
- **Integration:** multi-patient replay with resource contention; emergency overrides; SOP usage logging; QR analytics path.
- **Performance:** event processing latency, skill evaluation time, memory growth.

Metrics

- Acceptance rate, alert-fatigue proxy, clicks/seconds saved, per-encounter repeats, time-to-first-response after suggestion.

Audit (regression guard)

- `domain\_requirements\_audit()` checks: equipment/QR/reservation/status JSON/analytics; SOP search & usage; lingering/snacks; risk score functions; troponin serial/delta; anti-spam; CT/Cath/Endo follow-ups; ethics flag.

Extending with a new skill (pattern)

```
def skill_expedite_labs(state: WorkflowState):
    if state.deteriorating and state.pending_labs_count >= 2 and not state.on_antibiotics:
        return {"action": "EXPEDITE_LABS", "reason": "Deterioration with multiple pending labs"}
    return None

SKILLS.insert(0, skill_expedite_labs) # prepend to raise priority
```

Run order (Kaggle)

1. Bootstrap cell (creates CONFIG + dirs under `/kaggle/working`). 2. Core imports & state/skills/resource tracker. 3. Domain patches (this v4.1 content). 4. **Run** `domain_requirements_audit()` — expect **PASS**. 5. Simulate or replay events; inspect `metrics.json`, `emissions.json`.

Open gaps

- Patient status board UI; SOP checklist/flow renderer; richer analytics; external maintenance paging; end-to-end integration tests.